Chapter 5 — Advanced topics

Batch CSV cleaning is the baseline

Learning objectives

- Contrast Bayesian and deep imputation, describe sliding-window streaming preprocessing
- Explain how documentation supports explainable AI

Advanced imputation

- Bayesian imputation models missing values with uncertainty
- Autoencoder imputation, Example 5.71, learns nonlinear reconstructions for retail baskets
- Both beat naive means when patterns are complex but need more data and compute

Example 5.70 — Bayesian imputation with clinical priors

- Example 5.70 — hands-on module
- The uncertainty-aware clinical case
- Explore the chapter example module
- View files: `modules/chapter5/example70/`

Real-time and streaming preprocessing

- Streaming finance, IoT, and vitals demand low-latency transforms
- Challenges include latency, arriving corruption, and scale
- Sliding windows, Example 5.72 on prices
- Batch assumptions fail when data never stops

Example 5.72 — Rolling window for streaming prices

- Example 5.72 — hands-on module
- Example 5.72 maintains a rolling feature window for aggregates
- Explore the chapter example module
- View files: `modules/chapter5/example72/`

Automated pipelines and explainability

- Automated pipelines orchestrate cleaning stages across storage systems
- For explainable AI

Example 5.75 — Document imputation for auditable models

- Example 5.75 — hands-on module
- Example 5.75 ties preprocessing logs to model cards
- Explore the chapter example module
- View files: `modules/chapter5/example75/`

Looking ahead

- Chapter 6 continues with exploration that detects many defects this chapter repairs
- Chapter 7 connects preprocessing to fairness
- Strong cleaning documentation makes both downstream chapters more trustworthy

Takeaways

- Advanced methods extend, not replace, core taxonomy and tools
- Streaming and explainability impose new constraints on familiar transforms
- Document every stage for production and audit

Next

- Complete the quiz for this part
- Pause for the quiz